Frans Matome Ngwepe
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PROFILE LINKS

LinkedIn – https://www.linkedin.com/in/frans-matome-ngwepe-b474522b1/
GitHub – https://github.com/tshidisonm
Coursera – https://www.coursera.org/learner/frans-matome-ngwepe-4394
Personal Website – https://tshidisonm.github.io/personal_website/
Microsoft Applied Skills Transcript – Transcript

SUMMARY

Frans Matome Ngwepe is a technology-focused graduate from Gauteng, South Africa. He studied Mathematical Sciences and Computer Science at the University of Limpopo, finishing his undergraduate degree in 2024 and his Honours degree in Computer Science in 2025. He performed well academically, with distinctions in key modules. He has practical experience from building machine learning and software projects, including a model that predicts high-risk pregnancies and a mobile app that uses AI and smart hardware to help manage livestock. Frans has also worked as a freelance tutor in mathematics and computer science and received mentorship in software development. He holds an Oracle AI Foundations certification.

EXPERIENCE

IT Lecturer
Institution: Richfield Graduate Institute of technology
Polokwane Campus
03 February 2026 – current
Modules Lecturing: *Software Engineering, Machine Learning, Data Manipulation and Visualization, Software Development, Computer Architecture, Cloud Computing*
Skills I am currently learning: *Plotly, Plotly dash, Seaborn, Cloud computing fundamentals*

Software Developer
Company: African Accounting and Tax consultants
01 June 2025 – current

- Developed a full-stack customer and company management system for storing client records, business entities, and supporting documentation.
- Built using Node.js, Express.js, SQLite, sql.js, HTML, CSS, JavaScript, and Google Drive API.
- Implemented RESTful APIs for customer, company, document, statistics, backup, and restore operations.
- Designed document upload, preview, and deletion functionality with files stored as SQLite BLOBs.
- Added Progressive Web App (PWA) capabilities, including offline caching, service workers, install prompts, and application manifest configuration.
- Integrated secure OAuth 2.0 authentication with Google Drive to enable automated cloud backup and restore of the application database.
- Developed backup status monitoring, token metadata display, and backup history interfaces for improved system administration.

Demonstrated skills in full-stack development, database design, cloud integration, offline-first application architecture, and API development.

AI/ML Graduate Internship
Company: Linkfields innovations
05 January 2026 – 30 March 2026

AI-Powered Web Data Extraction & Chatbot Development
- Developed automated web scraping pipelines to extract and structure data from company websites.
- Designed and implemented an interactive chatbot to improve user engagement and information access.
- Demonstrated skills in data acquisition, NLP integration, and conversational AI development.

RAG-Based Intelligent Report Generator
- Built a Retrieval-Augmented Generation (RAG) system to generate context-aware, data-driven reports.
- Integrated vector-based retrieval mechanisms with large language models to enhance output accuracy.
- Applied machine learning, prompt engineering, and intelligent document generation techniques.

FinSight Prototype Development
- Developed a FinSight prototype to demonstrate AI-driven analytics capabilities for client presentations.
- Translated business requirements into a functional proof-of-concept solution.
- Showcased strengths in data analysis, solution architecture, and technical product pitching.

Automated Audit Analytics & IT Governance Project
Company: Accountants Advisory
01 February 2026 – 03 February 2026

- Developed an automated audit analytics toolkit to identify procurement control failures and business process risks
- A practical assessment for evaluating business processes, IT risks and controls, and Python-based data analytics across an end-to-end procurement and inventory lifecycle.
- Tools: Python, CSV data analysis, Risk & Control Matrix (RCM), audit analytics techniques.

Outcomes:
- Automated inventory reconciliation across thousands of records
- detected self-approvals and high-value variances
- designed a 10-point RCM covering access control and data integrity to reduce financial risk

Freelance Computer Science Tutor | Tutorship content and Testimonials - https://github.com/tshidisonm/Tutoring
Jan 2024 – Nov 2025
- Tutored Final-level Computer Sciences undergraduate students in Operating Systems, Computer Networks, Artificial Intelligence, and Databases, Mathematical Analysis, Abstract Algebra, and Complex Analysis.
- Provided academic support, practical guidance, and assignment assistance.
- Balanced tutoring responsibilities with Honours research and coursework.
- Delivered both in-person and online sessions alongside final-year studies.
- Supported students with exam preparation, problem-solving, and concept clarification.

Software Development Mentee (Mentored Project)
Jan 2024 – Nov 2024
Mentor: Jan Botha (Site Reliability Engineer)

- Completed a full-stack software project (Livestock Management Solution) and achieved a final score of 80%.
- Built a mobile app using React Native and Expo with a Supabase backend.
- Connected the app to hardware devices using Bluetooth Low Energy, including an ESP32 scale and RFID reader.
- Created an AI chat feature that queried databases using SQL and integrated with Google Gemini APIs.
- Developed Python APIs to connect system components and managed code using Git and GitHub.
- Solved complex hardware and software integration problems.
- Recognized by the mentor for strong planning, research skills, and a high standard of work.

Technical Projects (Academic & Independent)
2023 – 2026

Handwritten Digit Recognition using Neural Networks | Java, DeepLearning4J, ND4J, Maven:
From May 2026 to June 2026

- Developed a feed-forward neural network in Java to classify handwritten digits (0–9) using the MNIST dataset containing 70,000 images.
- Designed a multi-layer architecture with 784 input neurons, hidden layers with ReLU activation, and a Softmax output layer for multi-class classification.
- Preprocessed image data through normalization and one-hot encoding to improve training performance.
- Trained and optimized the model using the Adam optimizer and cross-entropy loss, achieving over 95% classification accuracy.
- Evaluated model performance using accuracy, precision, recall, and confusion matrix metrics.
- Implemented the solution using DeepLearning4J and ND4J, with Maven for dependency management and JUnit for testing.
- Built a prediction module capable of performing real-time digit classification from input images.
- Demonstrated skills in Java development, deep learning, data preprocessing, model evaluation, and machine learning system design.

Automated Audit Analytics & IT Governance Project: Developed a Python-based automated audit analytics toolkit to assess procurement controls and IT risks across an end-to-end procurement and inventory lifecycle. Automated inventory reconciliations, detected control breaches, and designed a 10point Risk & Control Matrix to reduce financial risk.

Livestock Management Mobile Application: Developed a React Native app integrated with ESP32 hardware, Bluetooth Low Energy, and an AI-driven chat interface querying a Supabase (SQL) database.

Service Marketplace Web Application: Built a full-stack web application using React, Vite, and Supabase to enable small businesses to list their services and user access.

Academic Research Project
Title: Early Prediction of High-Risk Pregnancies using Machine Learning
Research summary:
Built a stacked ensemble (XGBoost, LightGBM, CatBoost + XGBoost meta-learner) with stratified K-fold CV and SMOTE to handle class imbalance; achieved 98.01% overall accuracy and strong high-risk detection, demonstrating strong generalization and clinical utility for decision support in low- resource settings.

EDUCATION

Master of Science Computer Sciences
Research Topic: A Generative And Adaptive Deep Learning Framework For Explainable And Data-Efficient Wi-Fi Fingerprinting.
University of Limpopo | January 2026 – Current

Bachelor of Science Honors Computer Sciences
Distinctions in: Advanced Databases, Speech and Language processing (NLP)
University of Limpopo | January 2025 – December 2025 | Average: 72%

Bachelor of Science Mathematical Sciences (Mathematics and Computer Sciences)
Distinctions in: Abstract Algebra, Mathematical Analysis, Complex Analysis, Introduction to Databases
University of Limpopo | January 2021 – December 2024 | Average: 74%

SKILLS

- Technical Skills: RESTful API calls, PostgreSQL, Supabase Database integration, Python for large language model functionality, data modelling, Database Design, Microsoft SQL, JavaScript and TypeScript, Java Spring boot.
- Development Skills: Frontend GUI design using React, React Native with EXPO and managing code via Git and GitHub.
- Systems Integration Skills: Connecting hardware components with software services as well as developing middle layer integration API's. Postman for API testing.
- Problem-Solving and Innovation: Integrating advanced technologies and solving challenges related to hardware and software communication.
- Productivity tools: Claude Code, Qodo, VS Code, Cursor Ai

CERTIFICATIONS

Microsoft Applied Skills: Build a natural language processing solution with Azure AI Language
Verify at – Microsoft Credential

Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate – This certification provides an introduction to AI and ML fundamentals, focusing on their practical use within Oracle Cloud Infrastructure.
Issued by: Oracle Verify at – Oracle Credential

Introduction to Modern Data Engineering with Snowflake - an online non-credit course authorized by Snowflake and offered through Coursera.
Issued by: Coursera Verify at – Coursera credential

Intro to Snowflake for Devs, Data Scientists, Data Engineers - an online non-credit course authorized by Snowflake and offered through Coursera.
Issued by: Coursera Verify at – Coursera credential

REFERENCES

1. Mokoele MS – Honours Research Supervisor 2025
   Lecturer at University of Limpopo
   Email: motlatso.mokoele@ul.ac.za Mobile: 072 105 3745 Alternative: 065 842 5279
2. Jan Botha – Software Development mentor 2024

Site Reliability Engineer
Email: jannemanbrandt@gmail.com Mobile: 072 556 8643

ACHIEVEMENTS AND MORE INFORMATION

- Bronze Standard of the Duke of Edinburgh's International Award - 2016
- Languages – English, Sepedi, Setswana, IsiZulu
- Driving License Category C1